Preprocessing and EDA

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Overview

This report details the second milestone phase of data processing for the "Messy Mashup" dataset. The analysis focuses on efficient metadata extraction, parallelized acoustic feature computation using `librosa`, and baseline model evaluation utilizing a Decision Tree classifier.

Question-Wise Analysis

Q1. What is the mean duration (in seconds) of the Jazz genre stems in the train (genres_stems) dataset?
Approach: Leveraged `soundfile.info()` to parse audio file headers rapidly, bypassing the computational overhead of decoding the full audio arrays.
Answer: 30

Q2. What are the unique sample rates present in the entire dataset?
Approach: Scanned headers across all genre stems, noise data, and mashups using the optimized `soundfile` library.
Answer: [22050, 44100]

Q3. How many empty or zero-byte audio files are present in the train dataset?
Approach: Traversed the file tree and used `os.path.getsize() == 0` to identify completely empty or corrupted audio files.
Answer: 0

Q4. What is the average peak amplitude (in dB) for vocal stems in the train dataset?
Approach: Converted the stereo arrays to mono, extracted the absolute peak, and computed the Peak Decibels using the formula: $Peak_{dB} = 20\log_{10}(\max(|y|)+\epsilon)$. Evaluated concurrently using `ThreadPoolExecutor`.
Answer: -12.49

Q5. What is the mean spectral centroid for the 'blues' genre in the train dataset?
Approach: Loaded stems with `librosa` at a fixed sample rate and extracted the mean of `librosa.feature.spectral_centroid`.
Answer: 1579.84

Q6. Which genre in the train dataset has the highest mean spectral centroid?
Approach: Aggregated the spectral centroid computations for all 10 genres and determined the global maximum.
Answer: hiphop

Q7. How many stem audio files in the train dataset contain silence in the first 0.5 seconds? (amplitude $<$ 1e-4)
Approach: Optimized the loading step by strictly loading only the first 0.5 seconds (`duration=0.5`). Flagged files where the absolute maximum amplitude remained below the 10^{-4} threshold.
Answer: 333

Baseline Modeling (Decision Tree Classifier)

For the following questions, a feature set comprising *Tempo, Spectral Centroid, Zero-Crossing Rate (ZCR), and Spectral Rolloff* was extracted from 10-second snippets. The data was split (80/20) and modeled using a `DecisionTreeClassifier` with `max_depth=5`.

Q8. What is the validation macro F1 score computed using the baseline code?
Approach: Extracted predictions via `clf.predict(X_val)` and evaluated using Scikit-Learn's `f1_score` metric with `average='macro'`.
Answer: 0.15

Q9. What is the precision of hiphop?
Approach: Queried the generated `classification_report(y_val, y_pred, output_dict=True)` for the 'hiphop' class.
Answer: 0.25

Q10. What is the recall of pop?
Approach: Queried the classification report dictionary specifically for the 'pop' class recall metric.
Answer: 0.2

Q11. Enter model accuracy.
Approach: Computed the overall accuracy score by comparing predictions against ground truth labels (`np.mean(y_pred == y_val)`).
Answer: 0.19

Q12. Which genre has the highest true positives?
Approach: Computed the `confusion_matrix` and iterated over the diagonal elements (`cm[i, i]`) to find the maximum value.
Answer: metal

Q13. Which genre has the lowest false negatives?
Approach: Calculated false negatives by subtracting the True Positive (diagonal) element from the total row sum for each genre in the confusion matrix, finding the minimum.
Answer: metal